SPORTS VISION & TACTICAL MAPPING

Dance Formation & Floor-Contact Intelligence

Team technical review, complete implementation roadmap, grading evidence plan and oral-defense checklist

Team: Muneeb Shakoor · Tomislav Novosel · Viviana Pajic
Prepared: 12 November 2026 · Status: Planning and repository audit; not a claim of completed functionality

Objective: Deliver a reproducible, defensible MP4-to-2D stage-mapping system that satisfies every MVP criterion, investigates all three Extra Mile tracks, and provides compelling evidence for the 25/35/20/10/10 rubric. Teacher has verbally approved dance as the application domain.

Critical principle: A high grade is earned by reliable experiments, label quality, causal error analysis and live technical mastery—not by feature count. No particular score can be guaranteed.

Executive decision

Continue the existing repository; do not restart. Its CVAT protocol, YOLO experiments, floor-support model, tracking investigations and saved comparisons are valuable. Freeze the original experiments and create a clean, separately testable application pipeline. Finish the rubric-critical detector evaluation and integrated dashboard before extending features.

Primary product claim: formation and spatial-coordination decision support. Do not claim objective choreography synchronization or judge replacement without independent pose/timing ground truth and validation.

1 | Rubric-to-evidence contract

Criterion / weight	Mandatory evidence	Excellence / defense
Data and annotation / 25%	40–80 challenging initial frames; pretrained person detector; explicit gap; team label rules; blur/occlusion policy; split and dataset manifest	Human-reviewed 300-frame expansion; double-labeled agreement sample; examples of corrected labels; dataset audit and learning curve
Modeling and fine-tuning / 35%	YOLO fine-tune; mAP@50 and mAP@50:95; train/validation logs	Baseline vs tuned models, controlled augmentations and LR/freeze ablations, precision/recall, losses, per-scenario metrics, repeatability
Failure/error analysis / 20%	Concrete FP/FN examples, confusion causes and correction hypotheses	Error taxonomy, root-cause audit, before/after casebook, unsuccessful trials and bounded conclusions
Pipeline and homography / 10%	MP4 → boxes → 4+ planar correspondences → cv2 homography → synchronized minimap → local Streamlit/Gradio	Reprojection error, calibration checks, contact uncertainty, video timing tests, usable UI
Process/GenAI ownership / 10%	Checkpoint history, feedback response, AI disclosure, individual code walkthrough	Decision log, reproducible runs, provenance, peer defense drills and honest limitations

Grade ceiling: Official brief says sound MVP with analysis can earn up to 85%; successfully defended Extra Mile work is required to access 85%+. Extensions are necessary for higher distinction, not sufficient.

2 | Existing repository: evidence-based review

Source: https://github.com/GCM-Novosel/dance-floor-contact (main branch inspected 12 Nov 2026). This is a static source/record audit, not an independently executed training run.

Keep / leverage	Audit or build next
Labeling_Guide.ipynb; 71-image A/B/C CVAT packs (60 train, 11 val), 7 confirmation images; instance IDs and visible/hidden/airborne/uncertain support states	Human review of all labels, independent agreement audit, broader 300-frame selection, temporal leakage audit
Research_Notebook.ipynb and experiments comparing box-bottom vs learned support, temporal variants, seeds and smoothing	Separate clean experiment summary, actual mAP@50 and mAP@50:95, held-out detector evaluation
YOLO detector training code, pretrained initialization, frozen/full-layer tuning, brightness variants	Controlled baseline and tuning matrix; store config, weights, metrics, hardware and failure cases
Tracking, trajectory fitting, camera-motion auditing and cached reproduction scripts	Full fresh inference on new MP4, identity metrics, calibrated stage mapping, UI and deployment instructions

Reported preliminary contact benchmark: on seven confirmation frames, 19/46 box-bottom, 25/46 earlier support and 29/46 expanded support predictions were within 3% of assumed mat width. These are saved results using draft labels from one choreography, not independent performance-level validation. The README explicitly states final end-to-end notebook follows human review/retraining.

Red flags: main notebook emphasizes contact accuracy rather than required mAP; 1.6 mat aspect ratio is assumed; a complete Streamlit/Gradio MP4 application was not identified; source code is experiment-heavy; cached repro is not fresh inference; no validated judge-quality sync metric.

3 | Product definition and non-goals

MVP user journey

- User uploads a static-camera MP4; app validates format, resolution, frame rate and clip length; logs errors rather than crashing.
- Detector outputs dancer boxes and confidence; tracker associates stable IDs; floor-support head estimates visible contact or abstains.
- Four or more non-collinear stage-plane correspondences define a homography; project contact points to normalized 2D stage coordinates.
- Display bounding-box video and synchronized minimap side by side; preserve timestamp/frame alignment; show unknown/airborne states explicitly.
- Export annotated video, per-frame CSV/JSON, calibration and experiment metadata; provide an offline sample demonstration.

Research-grade extensions

- Extra Mile 1: ByteTrack/BoT-SORT baseline and tracking trajectories; compare ID switches, fragmentations, coverage and smoothness. Kalman smoothing is a second controlled trial, not a substitute for correct identity.
- Extra Mile 2: automatic stage landmarks/keypoints; first determine whether visible, repeatable stage-floor anchors exist. Train/evaluate only if sufficient labels and meaningful benefit over manual calibration. Report

negative result if not viable.

- Extra Mile 3: stage occupancy KDE heatmaps, team/cluster formations, spacing and symmetry; dynamic jersey/color clustering is optional and likely weak if costumes are uniform.
- Additional research stretch: pose-based timing/synchronization with human-defined events and uncertainty. Treat as experimental and do not conflate position alignment with movement synchrony.

4 | Dataset and annotation protocol

Frame selection and splitting

- Start with the prescribed 40–80 challenging frames; use the 71-image review pack as an audited pilot, not an automatic final ground truth.
- Expand to 300 unique, informative frames after reviewing the pilot; record each frame’s source video, timestamp, action, pose, blur, lighting, occlusion and selection rationale.
- Sample across formation changes, standing/kneeling/seated/jumping, crossings, edge positions, low contrast and hard negatives. Avoid 300 near-duplicate adjacent frames.
- Prefer train/validation/test separation by independent performances; if only one performance exists, use blocked temporal partitions with guard gaps, perceptual duplicate checks and clear generalization caveats.
- Lock test partitions before model selection. No tuning against final confirmation; keep versions of every manifest and hash source files.

Annotation policy and quality assurance

- For each distinguishable dancer: one tight visible-extent bounding box; never fabricate fully hidden boxes. Assign image-local positive instance_id; record class, contact state and support type.
- Only visible contacts receive exactly one floor_support point linked to the same ID. Mark hidden, airborne and uncertain without a fabricated point. Define midpoint rules for two supports; distinguish feet, knees, seat, palms.
- Agree on partial visibility, blurred silhouettes, boundary clipping, overlapping dancers and impossible-to-decide cases with illustrated examples. Preserve excluded/uncertain cases for audit.
- Double-label 10–15% of frames independently, adjudicate disagreements and report box IoU, contact-state agreement and point-distance disagreement. Audit label completeness per frame.
- Use CVAT-assisted proposals only as drafts; record who reviewed each pack, date, number of corrections and unresolved ambiguity. Verify exports programmatically.

5 | Modeling: the 35% priority

Controlled experiment matrix

Experiment	Changed factor	Record
E0	Generic pretrained person detector, no fine-tune	mAP@50, mAP@50:95, P/R, latency, per-condition failures
E1	Fine-tuned YOLO on reviewed pilot	Same metrics on unchanged validation
E2	Fine-tune with expanded training data	Learning curve at e.g. 60/120/200/240 unique train frames where feasible
E3	Freeze vs unfreeze / LR schedule	Hold data and other settings fixed
E4	Blur, brightness, scale augmentation ablations	Performance by blur, occlusion and small subject strata
E5	Support point model vs box bottom vs pose landmark	Contact localization error, abstention, downstream map error
E6	Tracker/temporal and seed repeats	Identity and jitter metrics; mean/range over seeds where affordable

- Track precision–recall curves, AP at IoU .50 and averaged .50:.95, per-class AP if multiple classes, confidence thresholds, confusion examples, epoch losses and training time.
- Ensure mAP is computed using standard detector evaluation, not just fixed-threshold IoU matching or contact-point tolerance. State class definition clearly (dancer/person).
- Use consistent evaluation splits, resolution, IoU/NMS settings, hardware and seeds. Select model on validation; report final test once. Avoid overinterpreting tiny improvements.
- Log architecture/backbone, pretrained checkpoint, input size, augmentation settings, optimizer, learning rate, epochs, patience, batch size, class mapping and dataset hash.
- Analyze training/validation loss divergence and explain false positives, false negatives, missed floor-work, duplicate boxes and lighting failures in relation to annotation and augmentation.

6 | Homography, geometry and calibration

- Confirm static or near-static camera by background feature registration; if camera moves, stabilize or update mapping and document the additional assumption.
- Identify at least four physically coplanar, non-collinear points with known or consistently normalized stage-template positions; label them on a reference frame.
- Estimate H using cv2.findHomography (RANSAC if overdetermined), validate orientation, invertibility, conditioning and projected boundaries; save calibration JSON.
- Evaluate reprojection error on independent landmarks not used in fitting; report pixels and normalized-stage units, plus visual overlay. If no physical dimensions, do not report metres.
- Project observed floor support, not box center. For hidden/airborne contacts, abstain or show a distinctly styled predicted location with uncertainty; never present an estimate as measured truth.
- Test edge-of-stage cases, jumps, camera drift, missing anchor points and wrong point ordering; explain planar assumptions during defense.

7 | Tracking, analytics and synchronization

- Compare ByteTrack and BoT-SORT or document why only one is viable; use consistent detections. Measure ID switches, identity F1/IDF1 or HOTA if reliable ground-truth IDs exist, plus coverage and fragmentation.
- Store observed vs predicted track states; bound interpolation gaps and confidence. Compare raw, Kalman and offline smoother on localization and lag. Offline smoothing may use future frames and must be labeled non-real-time.
- Heatmaps: normalize for time visible and coverage; separate observed from imputed positions; disclose KDE bandwidth and boundary effects.
- Formation analytics: pairwise distances, center of mass, occupancy, spatial symmetry, formation transition timelines; avoid a normative score without choreography reference.
- Optional synchronization: establish human-reviewed timing/pose events and tolerance, compare against baseline, show counterexamples. Never equate spatial proximity with synchronized movement.

8 | Clean repository architecture

Recommended new structure (preserve old experiment history):

app/streamlit_app.py
src/{video_io,detection,tracking,floor_contact,calibration,projection,analytics,pipeline}.py
configs/{models,pipeline,calibration}.yaml
data/{manifests,annotations}/ (large raw data via Git LFS or external manifest)
experiments/{detector,support,tracking,homography,error_analysis}/
models/ (versioned weights or download manifest)
tests/{unit,integration,regression}/
docs/{annotation_policy,dataset_card,experiment_log,error_casebook,genai_disclosure,defense_notes}.md
scripts/{prepare_data,train,evaluate,run_demo}.py
requirements.txt · README.md · LICENSE/rights notes · .gitignore

- Do not silently relocate or erase historical scripts. Tag the current baseline and preserve experiment artifacts and Git history; add clean wrappers/modules around selected components.
- Make one command to run the app and one command to reproduce the selected evaluation. No absolute Windows paths, missing checkpoints or cached-only success claims.
- Pin dependencies, note Python/CUDA versions, provide CPU fallback and reproducible demo inputs. Keep private/third-party footage rights documented.
- Add CI checks for lint, import, unit tests, schema validation and a small synthetic/sanitized integration smoke test. Avoid CI dependence on GPU training.

9 | Milestone roadmap and exit gates

Gate	Deliverables and acceptance condition
M0 / scope	Teacher approval recorded; exact source footage, camera feasibility, four anchors, stage dimensions/normalization and measurable claim defined.
M1 / data	CVAT 71-frame human review complete; policy and inter-annotator audit; 300-frame plan and locked split manifest.
M2 / baseline	Pretrained detector metrics, box-bottom projection baseline, first synchronized 10–20s clip and documented failure gallery.
M3 / modeling	YOLO fine-tune and controlled ablations complete; mAP tables, losses, config, model selection and one-time held-out evaluation.
M4 / geometry	Homography calibrated; independent reprojection error; contact-point ablation; uncertainty/airborne handling.
M5 / integration	Fresh MP4 processed end-to-end on clean setup; side-by-side synchronized Streamlit/Gradio video and map; exports and tests.
M6 / extra miles	Tracking + trajectories defended; automated keypoint experiment assessed; spatial analytics and heatmaps verified; optional pose sync research.
M7 / evidence	Error casebook, checkpoint feedback, GenAI disclosure, experiment ledger, reproducibility instructions and all-member oral rehearsal.

Scheduling: Assign actual dates after confirming the deadline. Protect the MVP and evaluation gates before spending time on optional extensions.

10 | Error analysis and engineering evidence

Required error-casebook format

Case ID | source frame/time | ground truth | prediction | error type | root-cause hypothesis | related annotation decision | controlled change | before/after metric | conclusion | artifact path.

- Minimum useful strata: overlapping dancers, floor-work, airborne motion, blur, dark costumes/background, partial off-screen, ID switches, stage-edge projection and false contacts.
- Capture visual examples of both FP and FN. Distinguish detection errors from tracker association, contact estimation and homography calibration errors.
- Run at least one explicit hypothesis-driven correction cycle per major recurring failure. Record failed hypotheses honestly and never present qualitative improvements without matching quantitative evidence.
- Maintain checkpoint diary: date, feedback received, action, experimental evidence and changed decision.

11 | Interface, QA and reproducibility acceptance tests

- Upload a fresh MP4 with valid static camera; process without requiring precomputed tracks; generate synchronized annotated video and stage map; verify consistent frame count and timestamp alignment.
- Invalid format, missing model weights, corrupt frame, no detections and calibration failure should yield clear messages. Avoid silently using old cached results.
- Test empty stage, heavy occlusion, jump, crossing, low light, boundary positions, variable frame rate, nonstandard aspect ratio and short clips.

- Unit tests: coordinate transform, homography inversion, annotation schema, no train/test overlap, track-state logic and point visibility; integration tests for a short deterministic video.
- Measure runtime, FPS, peak RAM/VRAM and machine specs; explicitly distinguish CPU vs GPU performance and offline vs real-time processing.
- Document video licensing/permission and privacy implications of person tracking; use as a judge aid, not an autonomous fairness claim.

12 | Oral-defense playbook (all three members)

- Explain YOLO backbone/head at an appropriate level, transfer learning, class adaptation, augmentations, loss dynamics, optimizer, confidence/IoU thresholds, precision/recall and mAP@50 vs mAP@50:95.
- Show why 300 frames were chosen, which are truly independent, how CVAT assistance was corrected, and how annotation changes affected convergence or failure patterns.
- Draw the MP4 → detector → tracker → support head → homography → analytics → UI data flow; identify uncertainty and cached-vs-fresh boundaries.
- Derive conceptually why four non-collinear planar correspondences are needed; explain that jumps break planar floor-contact assumptions.
- Defend each extension with baseline, metrics, failure examples, trade-offs and a clear decision to keep/reject. Each teammate must navigate code and rerun a small test unaided.
- Practice adversarial questions: What is your most misleading metric? How would the model fail on a new troupe? What did AI write? What did you verify? Which extension was not worth its cost?

13 | Team workflow, ownership and GenAI disclosure

- Shared success does not remove individual defense responsibility. Assign a primary owner and secondary reviewer for each subsystem, then rotate code walkthroughs.
- Use issues/PRs for changes; require a short rationale, test evidence and result link. Avoid concurrent unreviewed edits to the same notebook.
- Record GenAI tools used, prompts/tasks at a meaningful level, generated code/ideas, human verification, edits and failures. Follow the university's exact GenAI policy if it specifies a format.
- Keep experimental claims traceable to versioned data, scripts and result files. Avoid claiming independent reproduction when using frozen detector or support caches.

14 | Final submission checklist

- [ ] Dance approval documented and scope accurately described.
- [ ] 40–80-frame targeted annotation MVP and reviewed 300-frame extension documented.
- [ ] Label policy, occlusion rules, inter-annotator audit, dataset card and leakage audit complete.
- [ ] Pretrained baseline and fine-tuned YOLO compared with mAP@50 and mAP@50:95.
- [ ] Augmentation/hyperparameter experiments, training curves and error-root-cause studies included.
- [ ] Four-plus keypoints, verified homography and synchronized MP4-to-minimap pipeline work.
- [ ] Streamlit/Gradio app runs on clean machine with documented environment and downloadable outputs.
- [ ] Tracking/trajectories, automatic-keypoint experiment, and tactical analytics each tested and defended (successful or negative result).

- [ ] Reproducible experiments, source footage permissions, performance and robustness tests documented.
- [ ] Checkpoint feedback, GenAI disclosure, team accountability and individual oral defense rehearsals complete.

15 | First 72 hours: practical kickoff

When	Action
Day 1	Freeze/tag existing main; meet and agree product claim, teacher approval note, video feasibility, four stage anchors and 300-frame annotation strategy.
Day 2	Each teammate reviews their CVAT pack; jointly adjudicate shared difficult examples; audit manifests and select final held-out blocks.
Day 3	Run pretrained YOLO baseline with official mAP metrics; build the simplest fresh-inference MP4 → boxes → manual homography → minimap demo; log failures.

Sources and scope

Primary specification: user-provided Computer Vision Challenge: Sports Vision & Tactical Mapping (MVP, Extra Mile, rubric and GenAI ownership). Repository: https://github.com/GCM-Novosel/dance-floor-contact ; inspected README, Labeling_Guide.ipynb, Research_Notebook.ipynb, CVAT schema/checklist, detector scripts, camera audit, final confirmation and reproduction protocols on 12 Nov 2026.

This document is a recommended engineering plan, not a statement that proposed tasks are complete. Reported experiment values are repository-authored and require independent team verification. No guarantee of maximum marks is implied.